

Online Appendix to “How much productivity do we need to reduce the workweek?”

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This online appendix collects the data sources, the partial-equilibrium omissions, the calibration details, the sensitivity grids, the sectoral decomposition, the type-level welfare table, the output decomposition, the historical TFP table, the derivations, and the asymmetric- $e(h)$ robustness referenced in the main paper. Section, equation, table, and figure numbers are prefixed with “A.”. Cross-references to the main paper resolve to the labels in the main document.

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A Data sources

All data are from publicly available sources. PNAD Contínua (IBGE/SIDRA), tables 4093, 4097, 6374, and 6413 (quarterly, 2023–2025), provides the informality rate, position in occupation, average hours, firm-size distribution, and the CES formal weight $\omega = 0.622$ (1 minus the narrow PNAD informality rate averaged over the four quarters of 2025). RAIS/CAGED (Ministry of Labor) provides annual administrative records of formal employment, used for firm-size distribution and formal/informal employment shares. The Penn World Table 10.01 provides the labor share for Brazil ($\text{labsh} = 0.578$), adjusted via Gollin (2002) to $\alpha = 0.35$. DIEESE (2024) provides the hours distribution across the formal workforce: $\theta = (0.085, 0.269, 0.646)$ for the $\{36, 40, 44\}$ hours bins. An auditing exercise in the replication package verifies that the non-targeted moments in the main text are not algebraic consequences of the calibration.

B What partial equilibrium omits

The static partial-equilibrium framework excludes seven general-equilibrium channels. Capital accumulation responds to the lower marginal product of capital after the effective labor contraction. Wages and prices adjust in general equilibrium, partially restoring formal labor demand. Labor force participation may rise with leisure. Monetary and fiscal authorities may accommodate. Workers may reallocate across sectors. The formalization decision may move in either direction. Work-sharing is shut down by the fixed- N assumption. All seven channels except the last have a negative expected sign on A_{req} .

The capital-adjustment bound in Equation (??) of the main paper uses a textbook user-cost elasticity $\varepsilon_K \in [0.2, 0.4]$ (Caballero, 1999) and the baseline capital share $\alpha = 0.35$. At these elasticities over a 3–5 year horizon, the effective A_{req} under the conservative specification falls from 8.18% to 6.4%–7.2%, and under the preferred specification to 4.8%–5.6%. The remaining six channels are individually small and of mixed sign. The headline numbers should be read as short-run upper bounds on the required TFP gain. The 1–2 year model horizon is not the same as the 30-year PWT series used to anchor A_{req} . The anchor is best read as the cumulative TFP gain required over a 3–10 year phase-in horizon.

C Calibration details

C.1 Hours-output elasticity and efficiency shape

The Pencavel evidence documents declining marginal product of hours at long workweeks for a specific industrial setting. Collewet and Sauermann (2017) find analogous long-hour productivity losses in a Dutch call-center panel. Bick et al. (2018) provide cross-country evidence consistent with declining marginal product of hours, though their primary focus is the relationship between hours and income. Neither paper identifies a precise peak location or an efficiency response below a peak. The choice to place the peak at $h^* = 40$ and to compare a preferred flat-below specification (no penalty for shorter hours) with a conservative symmetric specification (equal curvature either side) is a modeling choice loosely informed by this evidence rather than a direct estimate from it. Section J reports four alternative shapes.

C.2 Informal efficiency and adjustment costs

The informal efficiency $\eta_I = 0.40$ is set at the midpoint of the informal-to-formal wage ratio range ($1/3$ to $1/2$) reported by La Porta and Shleifer (2014), complemented by the Mexican estimates in Busso et al. (2012). Under competitive wage-taking the CES first-order conditions identify η_I through the wage (marginal-product) ratio rather than through value-added per employee, which conflates productivity differences with capital-per-worker gaps between formal and informal firms. The adjustment costs $\gamma_F^S = 0.12$ and $\gamma_F^L = 0.03$ reflect the effective marginal compliance cost of CLT/FGTS obligations for small and large firms. Sensitivity over $\gamma_F^S \in [0.08, 0.16]$ and $\gamma_F^L \in [0.01, 0.05]$ shifts A_{req} by less than ± 0.2 pp. The convex informality cost π_m is calibrated endogenously to ensure non-negative wedges at the informality target.

C.3 Reconciliation with higher substitution elasticities

Refugee-crisis studies report substitution elasticities between worker types of $\sigma \approx 11\text{--}30$ (Ottaviano and Peri, 2012; Bahar et al., 2025). These magnitudes are not directly comparable to σ_{sub} as defined here for three reasons. The margin differs. Refugee studies measure substitution between workers of different origin in a market-clearing equilibrium, while σ_{sub} captures intra-firm substitution between formal and informal employment regimes. The horizon differs. Refugee studies identify medium-run responses that include firm entry and organizational reallocation, while the model here holds the firm distribution fixed. The unit differs. Refugee studies typically include self-employed workers, while the aggregator concerns salaried employment. DSGE models of informality (Leyva and Urrutia, 2020; McKiernan, 2021) similarly use σ parameters that govern sectoral mobility rather

than intra-firm substitution. In Ulyssea (2018), formal and informal workers are effectively perfect substitutes within the firm, with productivity differences captured entirely by the distribution of firm types. The calibration $\sigma = 1.326$ with $\eta_I = 0.40$ can be read as a parametric decomposition in which the formal–informal wage ratio directly pins the per-efficiency-unit relative productivity of informal labor, and a modest intra-firm substitutability ($\sigma > 1$) absorbs the remaining reallocation margin, both jointly disciplined by the observed wage premium.

C.4 Disciplining σ_{sub} : full results

For each σ in a grid over $[0.40, 3.00]$, the full model is recalibrated and the implied formal–informal wage premium is computed from the CES first-order condition,

$$R(\sigma) = \frac{\omega}{1 - \omega} \cdot \left(\frac{L_F}{L_I} \right)^{\rho-1} \cdot z,$$

where $z = \text{eff}_{h_F}/(\eta_I h_I e(h_I))$. The mapping is monotone in σ . At the recalibrated $(\omega, \eta_I) = (0.622, 0.40)$, setting $R \in [1.15, 1.55]$ delivers $\sigma \in [1.116, 1.469]$ and the conservative midbox $A_{\text{req}} \in [7.50\%, 8.58\%]$ (the full (σ, ω, η_I) -box envelope is $[6.93\%, 9.23\%]$). Derenoncourt et al. (2025) estimate a small formal-to-informal reallocation elasticity of about -0.28 from the Brazilian minimum-wage passthrough, consistent with the modest intra-firm σ used in this paper.

D Sensitivity

Table A.1 reports A_{req} under univariate variation of each parameter within economically-disciplined ranges. Three parameters are load-bearing: σ , ω , η_I . The remaining primitives shift A_{req} by at most ± 0.3 pp. Table A.2 reports the full (σ, ω) grid. The qualitative finding that A_{req} exceeds 4% across all combinations in the joint grid is fully robust.

Table A.1: Univariate sensitivity within economically-disciplined ranges (conservative).

Parameter	Disciplined range	A_{req} range	Hierarchy
σ_{sub}	[1.116, 1.469]	[7.50%, 8.58%]	High (load-bearing)
ω	[0.58, 0.66]	[7.65%, 8.65%]	High
η_I	[0.33, 0.50]	[8.02%, 8.32%]	Moderate
α	[0.30, 0.40]	[7.3%, 8.5%]	Moderate
e_q	[0.40, 0.80]	[7.8%, 8.1%]	Low
h_I	[38, 46]	[7.9%, 8.1%]	Low
ν	[1.0, 3.0]	ΔCV : [-3.4%, -3.3%]	Low (welfare only)

Notes: Each row varies one parameter inside its data-consistent range and holds all others at baseline. “Disciplined range” is the interval drawn from external evidence (Section C). “ A_{req} range” is the implied interval for the required TFP gain under the conservative $e(h)$ specification at the $44 \rightarrow 36$ h cap. “Hierarchy” classifies each parameter by the width of its A_{req} interval.

Table A.2: A_{req} (%) by σ and ω at $\eta_I = 0.40$ (conservative).

$\sigma \setminus \omega$	0.58	0.622	0.66
1.116	7.00	7.50	7.96
1.326	7.65	8.18	8.65
1.469	8.03	8.58	9.06

Notes: Each cell is the required one-off TFP gain (%) at the $44 \rightarrow 36$ h cap under the conservative symmetric $e(h)$ specification, for the listed (σ, ω) pair at $\eta_I = 0.40$ and all other parameters at baseline. The bolded row $\sigma = 1.326$ is the calibrated central value (corresponding to $R \approx 1.40$, the midpoint of the MNR window). The MNR-consistent interval at the recalibrated $(\omega, \eta_I) = (0.622, 0.40)$ is [1.116, 1.469]. Over the three-way (σ, ω, η_I) box with $\eta_I \in \{0.33, 0.40, 0.50\}$, the envelope across the eight corners is [6.93%, 9.23%].

E Sectoral decomposition

This section applies the national framework separately to three broad sectors (agriculture, industry, services) with sector-specific parameters drawn from PNAD Contínua Q4 2024 microdata (209,311 occupied persons, survey-weighted). Common structural parameters (α , σ_{sub} , ω , κ , h^*) are preserved from the national calibration. Each sector is treated as an independent instance of the national model. The exercise does not model cross-sector mobility or input-output linkages.

Table A.3 reports sector-specific parameters and the calibrated wedges. Agriculture’s informality rate is 71.8%, nearly twice the national average of 44.2% under the broad IBGE definition. Industry (44.4%) and services (41.2%) are closer to the mean. The calibrated wedges range from $\tau_s = 1.48$ in services to $\tau_s = 5.63$ in agriculture. The agricultural wedge is benchmarked against FUNRURAL-SENAR-INSS employer-side obligations of 30–35% of wages plus seasonality-related enforcement costs (Lemos and Galvão, 2020). The magnitude is large and not implausible given high institutional friction and a small formal base.

Table A.3: Sector-specific parameters and calibrated wedges.

Sector	λ_s	VAB	Inf. rate	θ_{36}	θ_{40}	θ_{44}	Hrs (F)	τ_s
Agriculture	0.076	0.077	71.8%	0.101	0.269	0.630	44.1	5.63
Industry	0.205	0.259	44.4%	0.048	0.417	0.535	42.5	2.10
Services	0.720	0.665	41.2%	0.171	0.410	0.419	40.8	1.48
National	1.000	1.000	44.2%	0.143	0.406	0.451	41.2	—

λ_s : employment share, PNAD Q4 2024. VAB: value-added share, IBGE Contas Regionais 2021. θ_b : share of formal workers in hours bin b . Hours: V4039 (habitual). Informality: broad IBGE definition (VD4009 + V4017 CNPJ).

Table A.4 reports the main sectoral results. The aggregate A_{req} imposes a uniform TFP multiplier on all three sectors and solves by bisection for the value that restores total output. Under the full reform, agriculture requires $A_{\text{req}} = 7.60\%$, industry 8.05%, and services 6.91%, with an aggregate of 7.21%. The aggregate is slightly below the national model’s conservative headline of 8.18% because this exercise uses PNAD habitual hours rather than DIEESE contracted hours. Industry’s A_{req} edges out agriculture’s. At $\sigma > 1$ and $\eta_I = 0.40$ the reallocation friction scales with the size of the formal base, which is larger in industry than in agriculture despite agriculture’s higher baseline informality. All three sectors see informality rise in response to the reform (agriculture +2.88 pp, industry +2.73 pp, services +2.10 pp), consistent with mildly substitutable formal–informal labor and the wage-disciplined informal efficiency parameter.

Services accounts for 71.0% of model-implied baseline output and contributes 4.8 of the 7.2 pp aggregate output loss. A sector-specific policy exempting agriculture from the cap would reduce the aggregate A_{req} by less than 0.2 pp. The decomposition treats each sector as an independent instance of the national model and abstracts from cross-sector

Table A.4: Sectoral results: required TFP gain and worker exposure.

Reform	Sector	A_{req} (%)	ΔY (%)	ΔInf (pp)	Affected (% of sector)	Burden ratio ($A_{\text{req}}/\text{Aff.}$)
44 → 40h	Agriculture	1.8	-2.4	+0.6	17.8	10.1
	Industry	1.6	-1.6	+0.4	29.7	5.4
	Services	1.3	-1.3	+0.4	24.7	5.2
	<i>Aggregate</i>	<i>1.4</i>	<i>-1.4</i>	<i>+0.4</i>	<i>25.2</i>	
44 → 36h	Agriculture	7.6	-9.4	+2.9	25.3	30.0
	Industry	8.0	-7.9	+2.7	52.9	15.2
	Services	6.9	-6.7	+2.1	48.7	14.2
	<i>Aggregate</i>	<i>7.2</i>	<i>-7.2</i>	<i>+2.3</i>	<i>47.8</i>	

Affected: formal workers above the new cap, as % of total sector employment. Burden ratio: A_{req} / affected share, descriptive. Aggregate A_{req} : uniform TFP gain solving $\sum_s Y_s(A(1 + A_{\text{req}}), h_1) = \sum_s Y_s(A, h_0)$ by bisection.

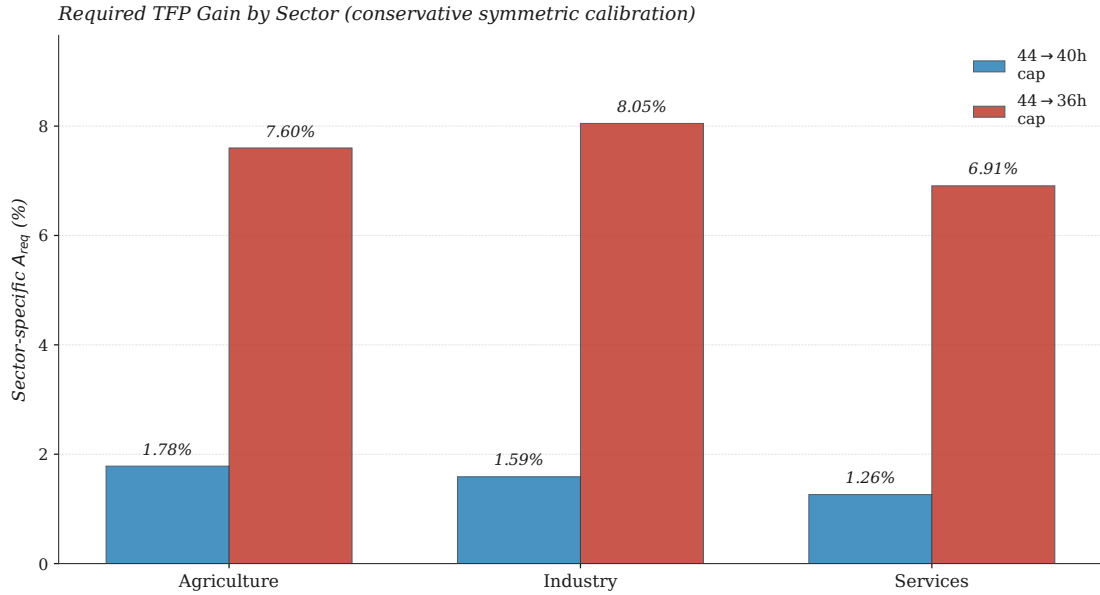


Figure A.1: Required TFP gain A_{req} by sector under the moderate 44 → 40h and full 44 → 36h reforms, conservative calibration.

Notes: Values are computed separately for each sector under the conservative symmetric $e(h)$ specification. Sector-specific parameters (informality, hours distribution, wedge τ_s) come from PNAD Contínua Q4 2024. Common structural parameters (α , σ_{sub} , ω , κ , h^*) are kept at their national values.

reallocation of labor and capital. It also collapses firm-size heterogeneity into a single representative firm per sector. A richer multi-sector framework with factor mobility is a natural extension.

F Welfare: type-level view

Table A.5 reports a type-level decomposition of the welfare effect at the $44 \rightarrow 36$ h reform under the conservative specification, using CES wage-bill shares. Workers who remain formal work approximately 6 fewer hours per week with a consumption loss of around 6% and a net welfare gain under GHH preferences with $\nu = 2$. Informal workers' welfare is largely unchanged because $h_I = 44$ is not subject to the cap. The mass of workers deformed at the margin faces sharp losses. At the moderate $\eta_I = 0.40$ this mass is material and smaller than under lower-efficiency specifications. The type-level figures are a first-order approximation. The exact representative-agent CV in Equation (??) of the main paper computes the composite at aggregate consumption and hours, and the gap between the share-weighted type-level sum and the aggregate CV is the Jensen-type correction implied by the concavity of the GHH composite.

Table A.5: Welfare by worker type ($44 \rightarrow 36$ h, conservative).

Worker type	ΔCV_k (first-order)	Share s_k
Formal stayers	+1.9%	60.1%
Informal (unaffected)	≈ 0	38.6%
Deformed	-32.0%	1.3%
Aggregate, Equation (??) of main	-3.63%	100%

Notes: ΔCV_k is the first-order compensating variation applied to type k (GHH, $\nu = 2$), evaluated at each type's consumption and hours. s_k is the CES wage-bill share used as an aggregation weight. The share-weighted sum of the type-level numbers is +0.009. The representative-agent value from Equation (??) of the main paper is -0.026. The gap is the Jensen-type correction from the concavity of the GHH composite.

Figure A.2 shows the welfare response as a function of an exogenous TFP gain at the 36h cap under the conservative specification. The welfare-neutral threshold is approximately 2.4% under the conservative specification and about 1.1% under the preferred specification.

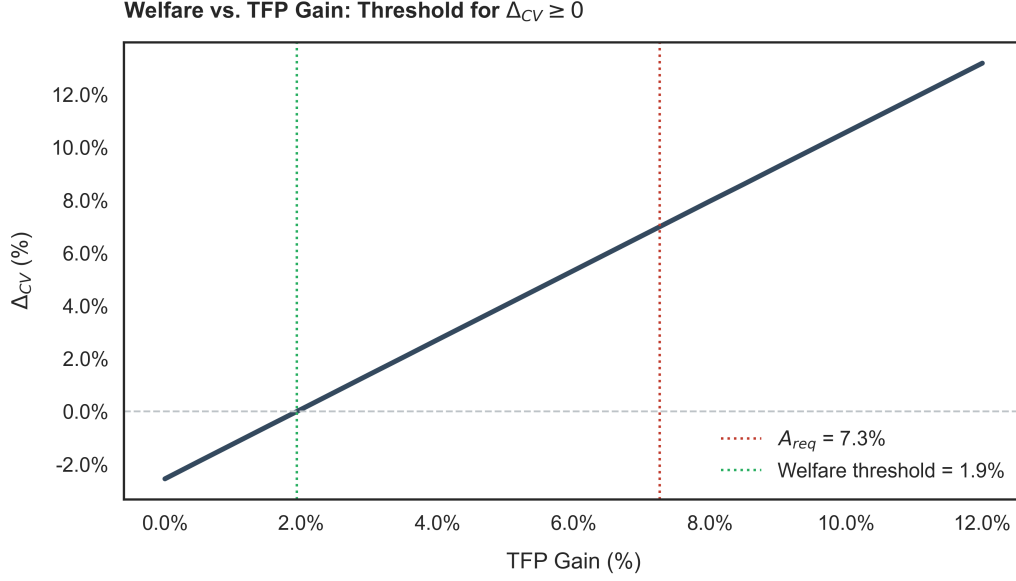


Figure A.2: Welfare ΔCV as a function of TFP gain at the 36h cap, conservative specification.

Notes: Horizontal axis: exogenous one-off TFP gain applied on top of the 44 $\rightarrow$ 36h reform. Vertical axis: aggregate welfare ΔCV from Equation (??) of the main paper. The curve is monotone-increasing in the TFP gain. The welfare-neutral threshold ($\Delta CV = 0$) is about 2.9% under the conservative specification and about 1.4% under the preferred one.

Table A.6: Welfare schedule, conservative specification.

Hours cap	A_{req}	ΔY	ΔCV	ΔInf
44	0.00%	0.00%	0.00%	0.00 pp
43	0.26%	-0.27%	+0.42%	+0.07 pp
42	0.66%	-0.70%	+0.63%	+0.16 pp
41	1.22%	-1.27%	+0.63%	+0.29 pp
40	1.92%	-2.00%	+0.41%	+0.46 pp
39	3.14%	-3.23%	-0.18%	+0.75 pp
38	4.58%	-4.65%	-1.06%	+1.09 pp
37	6.26%	-6.25%	-2.21%	+1.48 pp
36	8.18%	-8.02%	-3.63%	+1.92 pp
35	10.57%	-10.14%	-5.45%	+2.46 pp

Notes: All rows are model outcomes under the conservative symmetric $e(h)$ specification, measured relative to the 44h baseline. A_{req} is the required TFP gain for the given cap. ΔY is the aggregate output change. ΔCV is aggregate welfare from Equation (??) of the main paper. ΔInf is the change in aggregate informality. The bold row is the proposed 36h cap.

G Output decomposition

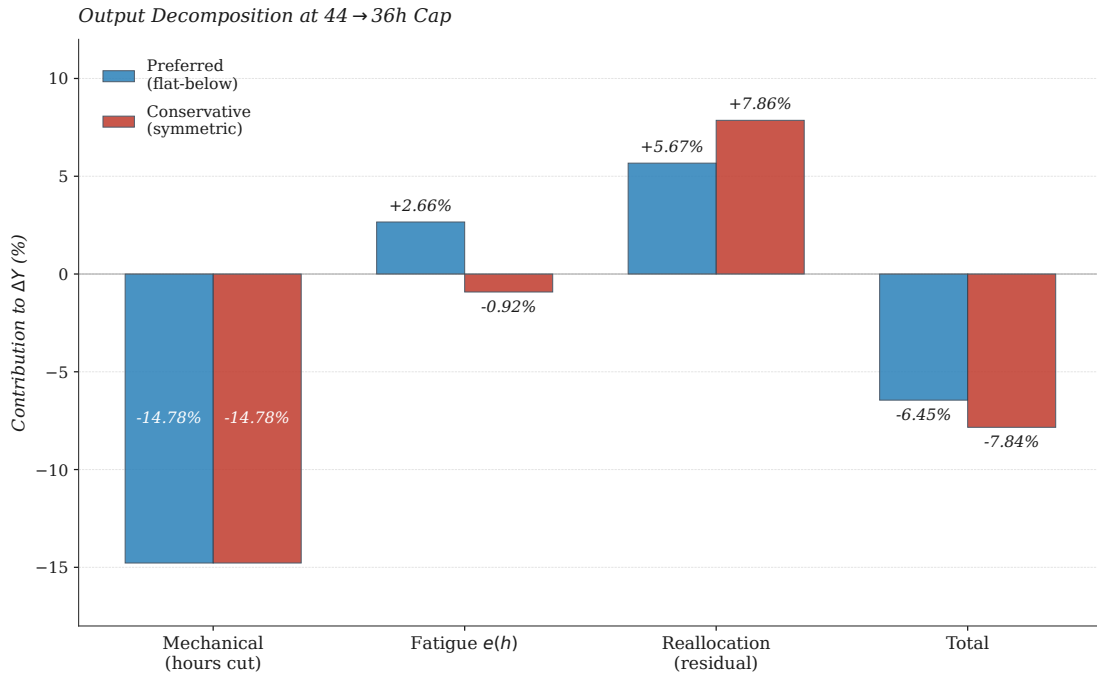


Figure A.3: Decomposition of ΔY at the 44 → 36h cap under each calibration.
Notes: The mechanical-hours channel (the direct effect of fewer contracted hours, holding per-hour efficiency fixed) is identical across specifications at -14.78% . The fatigue channel (the contribution of per-hour efficiency $e(h)$) switches sign between the preferred flat-below ($+2.39\%$) and conservative symmetric (-0.92%) specifications. The reallocation residual is the CES cross-term and is moderate in both cases because $\sigma > 1$ makes formal and informal labor imperfect substitutes and the moderate $\eta_I = 0.40$ (informal/formal wage-ratio midpoint) tempers but does not eliminate the reallocation cost when informality rises.

H Historical TFP table

Table A.7: Historical Brazilian TFP growth and years required to accumulate A_{req} .

Horizon	Span (yrs)	TFP growth (%/yr)	Years to A_{req}	Feasibility
1954-2019	65	+0.29	22.4	feasible
1954-1980	26	+2.29	2.8	feasible
1980-2000	20	-1.31	∞	<i>never</i>
1990-2019	29	-0.60	∞	<i>never</i>
1995-2019	24	-0.83	∞	<i>never</i>
2000-2019	19	-0.72	∞	<i>never</i>
2003-2013	10	+0.26	24.7	feasible
2010-2019	9	-1.88	∞	<i>never</i>

Source: Penn World Table 10.0, variable **rtfpna** (real TFP at constant national prices). Growth rate is annualized $(\text{TFP}_1/\text{TFP}_0)^{1/T} - 1$.
Years to $A_{\text{req}} = 6.63\%$ computed as $\ln(1 + A_{\text{req}})/\ln(1 + g)$.

I Derivations

I.1 κ from e_q

With $e(h) = \exp\{-\kappa(h - h^*)^2\}$, the hours-output elasticity at h_{ref} is

$$e_q = \left. \frac{d \ln(h \cdot e(h))}{d \ln h} \right|_{h=h_{\text{ref}}} = 1 - 2\kappa h_{\text{ref}}(h_{\text{ref}} - h^*),$$

so

$$\kappa = \frac{1 - e_q}{2h_{\text{ref}}(h_{\text{ref}} - h^*)}. \quad (\text{A.1})$$

With $e_q = 0.60$, $h_{\text{ref}} = 42.244$, and $h^* = 40$, $\kappa = 2.110 \times 10^{-3}$, so $e(36) = e(44) = 0.9668$ and $e(40) = 1$.

I.2 Firm first-order condition

The firm problem has

$$\frac{\partial Y_g}{\partial L_{F,g}} \cdot h_{F,\text{avg}} \cdot e_F \cdot \omega \cdot \left(\frac{L_{F,g}}{L_g} \right)^{\rho-1} = \tau_g + \gamma_{F,g}(N_{F,g} - \bar{N}_{F,g}) + \pi_m N_{I,g}, \quad (\text{A.2})$$

where $\partial Y_g / \partial L_{F,g} = (1 - \alpha)Y_g / L_g \cdot (\partial L_g / \partial L_{F,g})$ and $\partial L_g / \partial L_{F,g} = \omega(L_{F,g} / L_g)^{\rho-1}$.

I.3 A_{req} bisection monotonicity

A_{req} solves Equation (??) of the main paper. The Cobb-Douglas structure $Y_g = AK_g^\alpha L_g^{1-\alpha}$ is linear in A and the optimal $L_g(A)$ is weakly increasing in A , so $\partial \sum_g Y_g / \partial A > 0$ and bisection converges to a unique A_{req} .

J Asymmetric $e(h)$ robustness

Table A.8 reports A_{req} under four specifications for $e(h)$. The baseline symmetric $e(h) = \exp\{-\kappa(h-40)^2\}$ with $\kappa = 2.11 \times 10^{-3}$ is compared against three asymmetric alternatives that vary curvature below $h^* = 40$ while keeping curvature above h^* at κ .

Table A.8: A_{req} under asymmetric $e(h)$.

Scenario	Assumption below h^*	$e(36)$	A_{req}
Symmetric (conservative)	$\kappa_{\text{below}} = \kappa$	0.967	8.18%
Half-curvature below	$\kappa_{\text{below}} = 0.5\kappa$	0.983	7.40%
Double-curvature below	$\kappa_{\text{below}} = 2\kappa$	0.935	9.73%
Flat below (preferred)	$\kappa_{\text{below}} = 0$	1.000	6.63%

Notes: Curvature above $h^* = 40$ is fixed at $\kappa = 2.11 \times 10^{-3}$ in all four scenarios. Only the curvature below the peak varies. $e(36)$ is per-hour efficiency at the proposed 36h cap (the further $e(36)$ is from one, the larger the efficiency penalty). A_{req} is the required TFP gain at the $44 \rightarrow 36$ h cap under each scenario.

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